



A Meta-analysis of Attachment at Work

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Abstract

This meta-analysis aimed to confirm and clarify the relationships between attachment style and various workplace correlates, including job performance, burnout, personality, and job satisfaction ($K = 109$ independent samples, $N = 32,278$ participants). Results provided the strongest support for the relationships between attachment style and the Big Five personality traits, burnout, and job performance. Anxious attachment was also related to a host of other correlates, including job stress, turnover intentions, job satisfaction, and work engagement. Additionally, dominance analysis was used and found that attachment style had incremental validity beyond the Big Five in the prediction of job performance, job satisfaction, organizational commitment, and leader-member exchange. Finally, we examined meta-analytic path models in which attachment style impacted job performance, job satisfaction, organizational commitment, turnover intentions, and leader-member exchange through trust in supervisor. This indirect effect was supported for all correlates and for both anxious and avoidant attachment. Overall, the results supported the use of attachment styles as an important correlate with organizational variables. Limitations, implications, and areas for future research are discussed.

Keywords Attachment style · Job performance · Job satisfaction · Organizational commitment · Burnout · Meta-analysis

Introduction

Attachment theory, originally developed to describe patterns of relationships between infants and their caregivers (Bowlby, 1969), has become one of the most influential theories in psychology at large (Finkel & Simpson, 2015). Attachment is defined as the human propensity to seek out and develop close affectional bonds with others (Bowlby, 1969). This work has been expanded to address how attachment styles developed early in life impact adult relationships (e.g., Bartholomew, 1997; Mikulincer & Shaver, 2016) and, in recent decades, has also been used to examine

relationships at work. Following Hazan and Shaver's (1994) seminal paper on the applicability of attachment theory to workplace relationships, research has examined the impact of attachment styles on employees' relationships with their leaders and mentors (e.g., Allen et al., 2010; Boatwright et al., 2010; Keller, 2003; Luke et al., 2020; Thompson et al., 2021), coworkers (e.g., Luke et al., 2020), and organization (e.g., Scrima et al., 2015).

The past 30 years of research on attachment styles at work has highlighted the relevance of attachment theory to the prediction of organizational outcomes. For example, research has found significant relationships between attachment styles and job performance (Frazier et al., 2015; Wu et al., 2014), job satisfaction (Kafetsios et al., 2014; Towler & Stuhlmacher, 2013), burnout (Littman-Ovadia et al., 2013; Pines, 2004), feedback seeking and acceptance (Allen et al., 2010; Wu et al., 2014), and organizational commitment (Richards & Schat, 2011; Scrima et al., 2015), among others. Thus, the literature suggests that attachment style can be utilized to inform a host of organizational attitudes, behaviors, and other outcomes (Harms, 2011).

However, the literature still lacks clarity and consensus, making it difficult to develop nuanced insight. Although several qualitative reviews have attempted to organize this

Additional supplementary materials may be found here by searching on article title <https://osf.io/collections/jbp/discover>.

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literature (e.g., Harms, 2011; West, 2015; Yip et al., 2018), no quantitative synthesis of the empirical findings has been conducted. Such a synthesis would be critical for establishing precise estimates of effect sizes for relationships that have been frequently studied and show consistent results at the primary level. Additionally, there are other relationships that have received inconsistent support or conflicting findings in the literature. Thus, the first purpose of the current meta-analysis is to conduct a much-needed quantitative synthesis to provide an accurate depiction of the relationships between attachment style and various workplace variables and clarify inconsistent findings and highlight any remaining inconsistencies that warrant additional attention at the primary-study level.

Further, secondary uses of meta-analytic data offer unique opportunities to expand upon results and test additional hypotheses (Oh, 2020). This can be an important tool for theory building and testing, as meta-analytic data can be used to address theoretical questions that cannot be addressed at the primary level. For example, meta-analytic data can be used to examine the incremental validity of a construct beyond related constructs (Oh, 2020). In the current meta-analysis, we examine the incremental validity of attachment style beyond the Big Five personality traits (Goldberg, 1990) with the outcomes of job satisfaction, job performance, organizational commitment, and leader-member exchange (LMX), which have been shown in previous meta-analyses to be significantly related to the Big Five (Barrick & Mount, 1991; Choi et al., 2015; Dulebohn et al., 2012; Judge et al., 2002). Determining the incremental contribution of attachment styles beyond the Big Five will help establish the theoretical discriminability of attachment styles from other personality constructs, as well as provide practical guidance by determining avenues for improving workplace outcomes. Additionally, we combine our meta-analytic results with others to test meta-analytic structural equation models (SEM) in which attachment style impacts various workplace outcomes (e.g., job performance, organizational commitment) through its impact on an employee's trust in their supervisor. Trust has been proposed as the most direct correlate of attachment styles (Feeney & Noller, 1996), and trust in one's supervisor can impact a host of organizational outcomes (Dirks & Ferrin, 2002). Therefore, we use meta-analytic SEM to examine the indirect effect of attachment styles on several organizational outcomes through the mediator of trust in supervisor.

In sum, the purpose of the current meta-analysis is to (1) synthesize the findings to provide precise estimates of relationships that are frequently studied and generally consistent and clarify the findings for relationships that have received inconsistent results; (2) examine the incremental validity of attachment style beyond the Big Five in predicted job satisfaction, job performance, organizational commitment,

and LMX; and (3) test a meta-analytic path model in which attachment style impacts various workplace outcomes through trust in supervisor. Our contributions provide much-needed clarity to the area of attachment at work and can steer the literature in worthwhile directions.

Theoretical Background

Attachment Styles

Attachment theory suggests that the attachment behavioral system, an innate psychological system that motivates people to seek support from others, is activated when a physical, psychological, or interpersonal threat is perceived (Bowlby, 1969; Yip et al., 2018). An attachment style is a schema or script that a person develops that describes the nature of relationships and influences what a person expects from relationships and how they believe they should behave in relationships. This schema is formed through instances of seeking support from caregivers in early life, and the recognized patterns of responses received in these instances (Ainsworth et al., 1978). In response to consistent, sensitive responding from others, a secure attachment style will form. This style contains a positive view of both the self and others, meaning the person believes they are worthy of support, and others are willing and capable enough to provide this support (Collins, 1996). When others are consistently absent or unavailable, an avoidant attachment style will develop. This style includes a positive view of the self as worthy of support, but a negative view of others as being unreliable or incapable of providing support (Richards & Schat, 2011). Finally, when support is inconsistently given when sought out, an anxious attachment style will develop, which includes a negative view of the self as unworthy of support, but a positive view of others as capable of providing support (Ainsworth et al., 1978).

Recent work examining the construct of attachment has suggested that a two-dimensional structure of attachment, including the dimensions of anxiety and avoidance, is the most theoretically sound and has received the most support (Raby et al., 2021). That is, attachment anxiety can be high or low, and simultaneously attachment avoidance can be high or low. A case in which both anxiety and avoidance are low represents attachment security. A case when anxiety is high and avoidance is low is labeled “anxious attachment.” When anxiety is low and avoidance is high, this is labeled “avoidant–dismissive.” Finally, when both avoidance and anxiety are high, this is labeled “avoidant–fearful” (Bartholomew, 1997). Given this framework, many primary studies measure only anxious and avoidant attachment patterns and operationalize secure attachment as a low score on both dimensions. We adopt this perspective in the current

meta-analysis, focusing on anxious and avoidant attachment patterns in the main analyses and utilizing specific reports of secure attachment, when available, as a method for examining the discriminant validity of the anxious and avoidant relationships.

Attachment at Work

The attachment style formed in childhood creates a schema for later interpersonal relationships (Bartholomew, 1997) and has been shown to be moderately stable into adulthood unless stressful attachment-related life events, such as parental death, divorce, or abuse, shift the trajectory (Waters et al., 2000). Additionally, the experiences in relationships in adolescence and adulthood can continue to shape one's attachment style across one's life (Bowlby, 1969; Hazan & Shaver, 1987). According to attachment theory, as an individual ages and becomes more independent from their caregivers, other adult relationships start to fulfill the same functions as the early caregiver relationships (Mayseless & Popper, 2007; Mikulincer & Shaver, 2003). Workplace relationships are one type of adult relationship that can fulfill these functions, and thus the attachment behavioral system, and the attachment style developed to cope with this system activation, can be activated in a work context (Harms, 2011; Yip et al., 2017).

A steady stream of research has examined the impact of attachment styles on relationships at work, supporting that workplace relationships can also fulfill attachment functions (Luke et al., 2020). One of the most prominent topics of interest for this research is leader–follower dynamics (Harms, 2011). Follower attachment style can impact an employee's preferences and expectations for leadership (Boatwright et al., 2010), and an incongruence between these expectations and the leader's behavior can influence satisfaction with the leader and eventual turnover (Keller, 2003). Further, activation of the attachment system in interactions with a leader can result in attachment-relevant coping strategies, such as distancing from an unsupportive leader (avoidant) or seeking attention or affirmation from an inconsistent leader (anxious), which can be hindering to work performance (Hudson, 2013; Wu et al., 2014; Yip et al., 2018).

Attachment style can also impact the behavior of the leader. Securely attached individuals are more likely to emerge as natural leaders (Berson et al., 2006). Furthermore, securely attached leaders are more likely to utilize relational (Doverspike et al., 1997) and transformational leadership styles (Popper & Mayseless, 2003), and be able to delegate work to others (Johnston, 2000). Conversely, avoidant leaders are more likely to utilize a task-oriented leadership style and delegate less often than anxious and secure leaders (Johnston, 2000). Anxiously attached employees are the

least likely to be nominated by their peers to be a leader (Mikulincer & Florian, 1995), and anxious leaders are likely to be rated by their subordinates as having lower task efficacy (Davidovitz et al., 2007).

Research has also examined the influence of attachment styles on relationships with coworkers and the organization at large. Attachment insecurity can impact instrumental helping between coworkers (Geller & Bamberger, 2009), conflicts with colleagues, and preferences for working with others or alone (Hardy & Barkham, 1994). Additionally, securely attached employees show higher levels of organizational commitment (Scrima et al., 2015), job satisfaction (Jang et al., 2013), work engagement (Falvo et al., 2021), and lower levels of turnover intentions (Nae & Choi, 2022).

Given this theoretical background, we now turn to the three aims of the current meta-analysis and review relevant literature regarding these aims.

Aim 1: Synthesis and Clarification

The first aim of the current meta-analysis is to synthesize the findings regarding relationships between attachment and workplace correlates that are frequently studied and have received generally consistent results, as well as clarify results that have been inconsistent. The explored relationships are grouped into four categories: other individual differences, negative emotional states, job attitudes, and performance. The findings for these groups of relationships are described below.

Other Individual Differences

First, attachment styles are often examined in relation to other individual differences. Attachment scholars have cautioned against viewing attachment styles as a composite of personality, instead arguing for its being a unique individual difference beyond personality and other individual differences (Fraley & Shaver, 2008; Harms, 2011). As a unique individual difference, attachment style can offer distinctive insights into behavior and important predictive ability. As such, it is critical to examine its relationships with other individual differences that are often examined in workplace contexts, such as the Big Five traits and self-efficacy, to understand how attachment styles may overlap with these other individual differences, and what gaps in understanding attachment style can uniquely account for.

In this section, we include the other individual differences of the Big Five personality traits and self-efficacy. This category is somewhat broader than the others described below; however, some overarching expectations for these relationships can still be made. Generally, the underlying perceptions of the self and others as valuable that form the attachment styles (Ainsworth et al., 1978)

can be expected to relate to other individual differences. For instance, the negative views of others as untrustworthy and unreliable that occur in avoidant attachment have been proposed to relate to general negative affectivity and neuroticism (Richards & Schat, 2011) as those negative perceptions spread into a more general negative worldview. Additionally, the negative views of the self that occur in anxious attachment could impact self-efficacy if the views of the self as unworthy of support and consideration begin to impact views of one's capabilities and worth as well (Richards & Schat, 2011; Shaver & Brennan, 1992). We further elaborate on the theoretical reasoning and general findings for these relationships below.

The Big Five Personality One of the most common individual difference correlates examined in relation to attachment style is the Big Five personality traits (Goldberg, 1990). Insecure attachment (i.e., anxious and avoidant attachment) is consistently positively correlated with neuroticism, and this is theorized to be due to the lack of security, negative emotionality, and negative views of the self or others that characterizes these attachment patterns (Richards & Schat, 2011). Insecure attachment has consistently been negatively related to extraversion and agreeableness. For avoidant attachment, this relation is likely due to the negative view of others as untrustworthy (Feeney & Noller, 1996), which could impact how much they choose to interact with others and how effective those interactions are (Richards & Schat, 2011). For anxious attachment, this relation is theorized to exist due to the perception of the self as unworthy of attention and support, which may inhibit social interactions out of fear of rejection, and lead to disagreeable interactions that are too forward, eager, or intense (Feeney & Noller, 1996). Further, avoidant individuals are prone to suppressing negative emotions (Dozier & Kobak, 1992), while anxiously attached individuals react more strongly to negative emotions (Gillath et al., 2005), and thus both are expected to be low on openness to experience as a mechanism for avoiding negative emotions (Shaver & Brennan, 1992), which has also been consistently supported. Although no theoretical connection has been made between conscientiousness and the insecure attachment styles, several empirical studies have found a negative association between them (e.g., Gruda & Kafetsios, 2019; Richards & Schat, 2011; Van vianen et al., 2003).

Self-Efficacy Self-efficacy has been found to correlate negatively with the insecure attachment styles (Furness, 2020; Johnstone & Feeney, 2015; Richards & Schat, 2011). This relation is theorized to develop from the negative views of the self and others, as well as difficulties with emotion regulation that can cause negative events to be more impactful on

the individual's functioning compared to their peers (Richards & Schat, 2011; Shaver & Brennan, 1992).

Negative Emotional States

Next, attachment style is also often examined in relation to negative emotional states such as job stress and burnout. It is important to understand the linkages between attachment styles and negative emotional states given how impactful these negative emotional states can be on health and well-being and workplace correlates (Sonnentag & Frese, 2013). Insecure attachment can create additional stressors when anxious or avoidant schemas are triggered, and prompt worrying about relationships or over-investing in work (Hazan & Shaver, 1990; Vîrgă et al., 2019). We explore these relationships in more detail below.

Job Stress The increased negative emotionality associated with insecure attachment is also proposed to be related to increases in experienced job stress (Collins, 1996; Harms, 2011; Richards & Schat, 2011). Furthermore, insecurely attached individuals are more likely to have relationship difficulties at work and thus are less likely to utilize social support to cope with stressors (Harms, 2011; Hazan & Shaver, 1990). Additionally, due to fear of failure and rejection for poor performance (anxious) or working more in an attempt to avoid social interaction (avoidant), insecurely attached individuals are more likely to become engrossed in work and face spillover into their personal lives (Hardy & Barkham, 1994; Vîrgă et al., 2019). In line with this, empirical work has found positive relationships between anxious and avoidant attachment and experienced job stress (e.g., Lopez & Ramos, 2016).

Burnout Relatedly, anxious and avoidant attachment have been found to relate positively to burnout (e.g., Hiebler-Ragger et al., 2020). As mentioned, attachment can invoke worrying about relationships and anxiety about performance, which may lead to investing too much effort into work tasks in order to gain the approval of others (Hazan & Shaver, 1990; Vîrgă et al., 2019), and those with avoidant attachment patterns may invest too much in their work as a method for maintaining independence and avoiding interactions with others (Harms, 2011; Hazan & Shaver, 1990). This over-investment in work can thus increase chance of burnout (Vîrgă et al., 2019). Further, those with insecure attachment styles react more strongly to work stressors, as insecure attachment is associated with increased negative emotionality and decreased emotion regulation abilities (Richards & Schat, 2011; Ronen & Mikulincer, 2009). Thus, those with insecure attachment styles often report higher levels of burnout (Hardy & Barkham, 1994; Hazan & Shaver, 1990).

Job Attitudes

Next, several relationships are explored in the literature between attachment styles and job attitudes. This category contains several important workplace phenomena, including job satisfaction, trust in one's supervisor, leader–member exchange, organizational commitment, turnover intentions, and work engagement. Similar to the individual differences category, this category contains several constructs that are proposed to have differential relationships with attachment styles. However, some overarching expectations based on attachment theory can be made. First, the distrust of others and negative views of one's own self-worth that characterize insecure attachment can negatively impact attitudes that are relational in nature, such as trust in one's supervisor, leader–member exchange, and organizational commitment (Richards & Hackett, 2012). Second, for attitudes based on the job itself, such as overall job satisfaction, turnover intentions, and work engagement, difficulties stemming from attachment insecurity can make one's work more straining and more complicated without access to colleagues and their resources and can lead to preoccupation with negative aspects or challenges (Hazan & Shaver, 1990; Reizer, 2015). These factors are likely to negatively relate to job satisfaction and work engagement (Hazan & Shaver, 1990) and positively to turnover intentions (Richards & Schat, 2011). We explore each of these in more detail below.

Job Satisfaction Similarly, anxious and avoidant attachment have also been found to relate negatively to overall job satisfaction (e.g., Berlanda et al., 2019; Schirmer & Lopez, 2001). Those with insecure attachment styles may be more preoccupied with relationship issues and their attachment needs, which can lead to underestimations of their work achievements, pessimistic appraisals of themselves at work, and overall dissatisfaction with their job (Hazan & Shaver, 1990; Reizer, 2015).

Trust in Supervisor An individual's trust in their supervisor has been examined frequently as an outcome of insecure attachment, as trust of others can be seen as the most direct outcome of attachment (Feeney & Noller, 1996; Harms, 2011). As avoidant attachment is explicitly characterized as a view of others as untrustworthy and undependable, this attachment style is likely related to low trust in one's supervisor. Although anxious attachment is characterized by a positive view of others, it is also characterized by a negative view of the self, which may manifest in a belief that a supervisor may not act in a dependable way (e.g., respond quickly, provide support, utilize clear communication) because they do not view the individual as worthy or valuable enough to receive that type of treatment. This proposed negative relation between insecure attachment and trust in supervisor has

been consistently supported by the literature (e.g., Frazier et al., 2015; Gruda & Kafetsios, 2019; Richards & Hackett, 2012).

Leader–Member Exchange Leader–member exchange (LMX) has similarly been examined as a relational outcome of attachment style, given its foundation in trust between the leader and follower (Graen & Uhl-Bien, 1995). Attachment anxiety has been theorized to relate negatively to LMX, such that an anxious follower will enter a relationship with a leader not expecting them to provide consistent support. They will thus be more likely to “cling” to the supervisor and resist offers of independence, which is critical for LMX development (Dienesch & Liden, 1986). In response to this, leaders may react by withdrawing support and attention or distancing themselves, which would confirm the anxious follower's expectations and thwart growth and LMX development (Richards & Hackett, 2012). Similarly, an avoidant follower would enter a relationship with a leader expecting them not to be dependable, and thus would remain distant from the leader and tackle challenges and tasks alone. This lack of engagement from the follower hinders LMX development and can also prompt the leader to develop similar feelings of lack of respect, dependability, and loyalty regarding the follower, which can hinder LMX development from the leader's perspective (Dienesch & Liden, 1986).

Organizational Commitment Organizational commitment has been examined in relation to attachment style in about a dozen primary studies, but the conclusions from these studies vary widely. Although about half of these studies report moderate to large negative correlations between organizational commitment and insecure attachment (e.g., Chopik, 2015; Richards & Schat, 2011), several studies reported weak or insignificant correlations (e.g., Scrima et al., 2015) or even positive correlations (e.g., Lavy, 2014; Schusterschitz et al., 2011). Thus, after reviewing the literature, it is still unclear what the relation is between organizational commitment and insecure attachment, and a deeper look into this relationship is warranted.

Turnover Intentions Turnover intention has mostly been shown to be moderately to strongly positively related to anxious and avoidant attachment styles (e.g., Dahling & Librizzi, 2015; Richards & Schat, 2011; Tziner et al., 2014). Additionally, some studies have measured secure attachment specifically and found negative relationships (e.g., Banerjee-Batist & Reio, 2016; Nae & Choi, 2022), as expected given the insecure attachment results. However, several studies have found insignificant relationships between turnover intentions and all three attachment styles (e.g., Falvo et al., 2012; Reizer, 2019), and others have found negative relationships between turnover intentions and insecure

attachment (e.g., Jackson, 2008) or positive relationships with secure attachment (e.g., Rivera, 2020). Thus, we look deeper into the relationship between attachment style and turnover intentions.

Work Engagement Finally, work engagement has generally showed moderate negative relationships with anxious and avoidant attachment (e.g., Byrne et al., 2017; Haroush & Koslowsky, 2020). However, a few primary studies reported positive, weak, or insignificant relationships (e.g., Falvo et al., 2021; Rivera, 2020). Thus, this relationship deserves further investigation.

Performance

Finally, attachment styles are often studied in relation to performance outcomes, including job/task performance and contextual performance (i.e., organizational citizenship behavior). Generally, it is theorized that preoccupation with unmet attachment-related needs can distract the individual from their work and impact their performance (Hazan & Shaver, 1990; Richards & Schat, 2011). We explore task and contextual performance separately below.

Job Performance Job performance (i.e., task performance) has been theorized to be negatively related to anxious and avoidant attachment due to a preoccupation with unmet attachment needs or interpersonal troubles that can impact performance effectiveness (Hazan & Shaver, 1990; Vîrgă et al., 2019). Further, insecure attachment can also impact performance indirectly through its impact on job stress and burnout (Vîrgă et al., 2019). Various empirical studies have supported the negative correlation between insecure attachment and task performance (e.g., Desivilya et al., 2006; Kale, 2020; Richards & Schat, 2011; Vîrgă et al., 2019).

Organizational Citizenship Behavior Similarly, organizational citizenship behavior (OCB) has also been theorized to be negatively impacted by the preoccupation with unmet attachment needs. Those with insecure attachment styles are often absorbed in relationship troubles and meeting their job requirements (Hardy & Barkham, 1994; Hazan & Shaver, 1990; Mikulincer & Shaver, 2007), thus limiting their capacity or desire to engage in extra-role behavior that is not mandatory (Richards & Schat, 2011). This negative relationship has been supported in the literature (e.g., Desivilya et al., 2006; Richards & Schat, 2011).

Aim 2: Incremental Validity Beyond The Big Five

The second aim of this meta-analysis was to examine the incremental validity of attachment styles over the Big Five personality traits (Goldberg, 1990) in predicting job

satisfaction, job performance, and organizational commitment. Previous meta-analyses have established that personality is important for job performance (Barrick & Mount, 1991), job satisfaction (Judge et al., 2002), organizational commitment (Choi et al., 2015), and leader-member exchange (Dulebohn et al., 2012). In addition to confirming whether attachment style is important for these outcomes, as we address in Aim 1, it would also be useful to understand how attachment style potentially adds to our understanding of these workplace outcomes above and beyond what personality can provide. Attachment styles represent an individual difference primarily related to how one perceives and interacts with others (Feeney & Noller, 1996). It contains a perception of one's self-worth in relation to others, and a perception of others as reliable and trustworthy (Ainsworth et al., 1978). Although these perceptions may be correlated with the Big Five traits (e.g., perception of low self-worth may be associated with neuroticism), this individual difference is not captured or subsumed by the Big Five personality traits, and to view attachment style as a composite of personality is misguided (Fraley & Shaver, 2008; Harms, 2011). Further, several primary studies have ascertained that attachment styles hold predictive power over the Big Five traits, especially for outcomes that involve a relational aspect (e.g., Nettle & Shaver, 2006; Roisman et al., 2007). Thus, it is likely that attachment style offers unique predictive power for job satisfaction, job performance, organizational commitment, and LMX beyond the Big Five traits, but to what extent remains unclear. We address this question with Aim 2.

Aim 3: Attachment and Workplace Outcomes

Finally, the third aim of this meta-analysis is to expand upon the meta-analytic correlations to test several structural equation models. We examine whether attachment style impacts job performance, job satisfaction, organizational commitment, turnover intentions, and LMX through its impact on an employee's trust in their supervisor. As previously mentioned, trust is often viewed as the most direct outcome of attachment styles (Feeney & Noller, 1996; Harms, 2011), as securely attached individuals view others as trustworthy, dependable, and good (Fraley & Shaver, 2008). Attachment theory suggests that through repeated instances of being consistently (avoidance) or sporadically (anxious) let down by others, an adaptive distrust of others forms (Bowlby, 1969). Additionally, this distrust generalizes to others beyond the initiating individual, and one begins to expect all relationships to follow the same pattern (Feeney & Noller, 1996). Thus, one can enter into new relationships with a pre-established distrust, and the expectation of being let down can lead to preemptive behaviors to prevent being hurt, such as "keeping them at arm's length" (avoidance), or excessive people pleasing to prevent them from leaving

(anxious; Yip et al., 2017). These expectations and behaviors based in attachment-based mistrust are likely to appear in leader–follower relationships (Harms, 2011; Hudson, 2013), especially because the leader–follower relationship is akin to the parent–child relationship that likely formed the attachment style (Harms, 2011).

The implications of mistrust in one’s supervisor due to expectations based in attachment extend beyond the mistrust itself. Trust in one’s supervisor has been found to have important implications for workplace outcomes (Dirks & Ferrin, 2002), such as job performance and job attitudes. Distrusting one’s supervisor can impact job performance through two possible mechanisms. First, when a subordinate feels as though their supervisor cannot be trusted, they are likely to tread carefully and invest energy in “watching their backs,” which can divert attention away from performance (Mayer & Gavin, 2005). Second, a lack of trust can impact performance through a social exchange perspective (Blau, 1964; Emerson, 1976), such that subordinates will be motivated to reciprocate their leader’s behavior, and invest less effort into their work when they believe their leader is untrustworthy (Dirks & Ferrin, 2002).

Trust in one’s supervisor is also likely to impact several attitudinal outcomes. Because supervisors often have a direct impact on employees’ job experiences, such as their training, job responsibilities, and performance appraisal (Rich, 1997), having a low level of trust in one’s leader can be distressing. This can impact one’s job attitudes, such as decreased job satisfaction and organizational commitment, and increased turnover intentions (Dirks & Ferrin, 2002). Additionally, trust in one’s leader is likely related to LMX, as previously described. Trust is considered an integral component of LMX (Graen & Uhl-Bien, 1995). Thus, without trust between the parties, LMX is likely to remain at a low quality and not develop over time.

With this reasoning, we examine the mediating role of trust in supervisor in the relationship between attachment style and the outcomes of job performance, job satisfaction, organizational commitment, turnover intentions, and LMX.

Methodology

Literature Search

To identify studies for inclusion in our meta-analysis, the initial literature search was conducted in the spring of 2022 using EBSCOHost, Google Scholar, and ProQuest Dissertations & Theses. The following search terms and Boolean operators were used: (“attachment style” OR “attachment orientation” OR “attachment”) AND (“work*” OR “job” OR “leader*” OR “follower*” OR “team*” OR “employ*” OR “mentor*”). This returned 611 relevant papers. Additionally,

we conducted a backwards search for studies that cited Hazan and Shaver’s (1990) seminal paper on attachment at work. To locate this additional literature, we searched EBSCOhost, Google Scholar, ProQuest Dissertations & Theses, Web of Science, and ScienceDirect. This search resulted in 418 papers. We also searched for relevant papers cited in reviews and theoretical papers, which resulted in 29 papers. Finally, we sought unpublished data from authors whose published works were included in the literature search. This resulted in two additional papers. Altogether, these efforts resulted in 1060 papers.

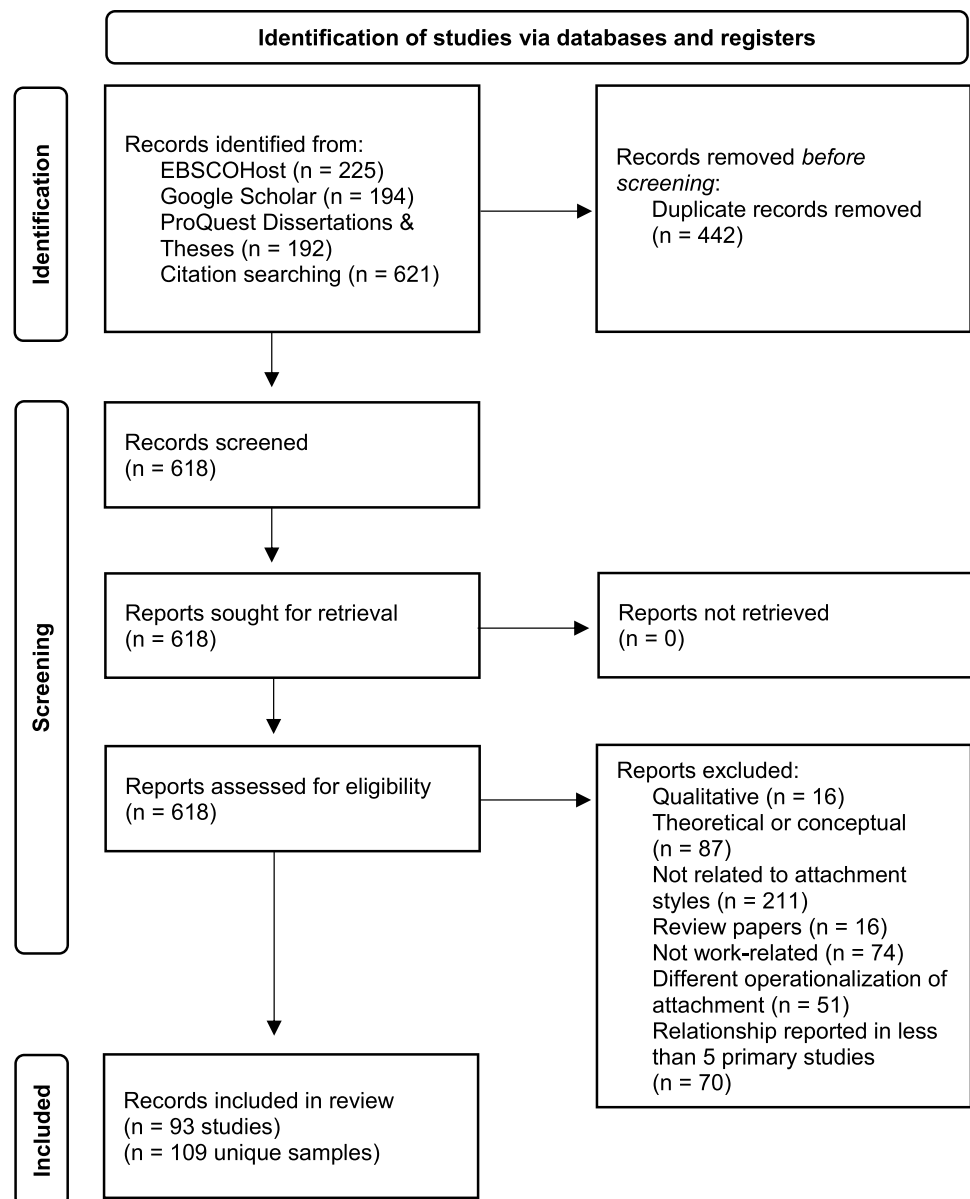
Inclusion Criteria

First, we excluded 442 duplicates within our list of work, which reduced the number of papers to 618. Next, we excluded any papers that were qualitative (16), theoretical or conceptual (87), not related to attachment styles (211), review papers (16), not work-related (74), or that measured or operationalized attachment in a way that was not aligned with the other papers (e.g., attachment to group, perception of leader as “secure base”; 51). After these exclusions, 163 primary studies remained. Finally, we excluded relationships that were reported in fewer than five primary studies, which reduced the final sample to 93 primary studies, and 109 unique samples. The literature search and selection process is shown in Fig. 1.

Coding Procedures

We utilized Hazan and Shaver’s (1990) framework for attachment styles, which encompasses anxious, avoidant, and secure attachment. Within this framework, these three attachment styles are treated as dimensions of overall attachment style, such that individuals receive a score for each dimension separately. Each dimension of attachment encompasses a specific pattern of relational behaviors, which provides a framework for understanding the impact of attachment on various outcomes. Thus, we coded relationships between these three attachment styles and the correlates. After exclusions to our database, only seven of the remaining studies utilized a four-style framework (i.e., anxious, avoidant-dismissive, avoidant-fearful, and secure) based on the two-dimensional structure of attachment. To incorporate these studies, we created a composite for avoidant attachment comprised of the avoidant-dismissive and avoidant-fearful styles. Although various measures of adult attachment exist that capture different subsets of the attachment styles (e.g., some measures only focus on avoidant and anxious attachment), measures generally capture attachment styles as continuous variables.

The coding of studies that met the inclusion criteria took place in the summer and fall of 2022. Two trained coders

Fig. 1 Identification and screening of studies

independently coded the necessary information from each primary study (e.g., correlations, reliability estimates, sample size). Disagreements were discussed in weekly calibration meetings until consensus was reached. A third coder served as a “tie-breaker” in cases where agreements were not reached via consensus.

Meta-analytic Procedures

Analyses were conducted using Hunter and Schmidt’s (2004) random-effects procedure in R with the *psychmeta* package (Dahlke & Wiernick, 2019). Effect sizes were sample size-weighted and corrected for measurement error. Additionally, we report the 80% credibility intervals and 95% confidence intervals for the corrected correlations. For studies that did

not report reliability estimates for a given construct, corrections were made for measure reliability using artifact distribution. Additionally, if only dimension-level relationships of correlates with attachment styles were included in a study that met our inclusion criteria, we combined several measures into indices. For example, dimensions of burnout (i.e., emotional exhaustion, depersonalization, and personal accomplishment) were aggregated to compute a composite burnout score via Hunter and Schmidt’s (2004) composite formula. As mentioned, meta-analytic estimates were not calculated for relationships with fewer than five samples, and we applied this rule to the incremental validity and structural equation modeling analyses for consistency. The homogeneity statistic, Q , was calculated to determine variation between studies. A significant Q statistic indicates that

between-sample moderators may be contributing to effect size differences between studies (Huedo-Medina et al., 2006).

The incremental validity of attachment style over the Big Five personality traits for job satisfaction, job performance, organizational commitment, and LMX (Aim 2) was examined using dominance analysis through the *domir* package in R (Luchman, 2023). Significant zero-order correlations between the attachment styles and the Big Five traits were confirmed before the variables were included in the dominance analysis. To conduct these analyses, we followed the practices proposed by (Viswesvaran & Ones, 1995), which includes four steps. First, the zero-order meta-analytic relationships between each of the three attachment styles, the three outcomes, and the Big Five traits were calculated in the main analyses. Additionally, zero-order meta-analytic relationships between the Big Five and the three outcomes were collected from other published meta-analyses. Next, a correlation matrix was created including correlations between the attachment styles, the Big Five traits, and the three outcomes (Viswesvaran & Ones, 1995). Finally, the correlation matrixes were used in the dominance analysis, which determines the relative importance of the predictors in the model based on each predictor's contribution to the overall model fit statistic (Luchman, 2023). The harmonic mean was used as the sample size for this analysis.

The proposed path models from attachment style to various workplace outcomes through trust in supervisor (Aim 3) was examined using meta-analytic SEM with the *lavaan* package in R (Rosseel, 2012). These analyses were conducted using a correlation matrix for all the included

variables. Secure attachment was not included as a predictor in these path models due to several relationships being represented by fewer than five samples. The current meta-analysis provided correlations between the attachment styles, trust in supervisor, and the five tested outcomes, as well as intercorrelations between the attachment styles. The meta-analysis by Dirks and Ferrin (2002) provided correlations between trust in supervisor and the five tested outcomes. The meta-analytic correlations between the tested outcomes were provided by the meta-analyses by Ng (2015), Riketta (2008), and Dulebohn et al. (2012). The harmonic mean was used as the sample size for this analysis.

Finally, we conducted sensitivity analyses to examine potential outlier and publication biases. To test for outliers, we utilized forest plots for each of the tested relationships to identify samples that appear to differ substantially from the rest of the samples. To test for publication bias, we created funnel plots for all the tested main-effect relationships.

Results

Aim 1

Tables 1, 2 and 3 summarize the results for overall relationships between attachment styles and the 18 tested correlates. The first aim of the meta-analysis was to provide synthesized effect sizes for relationships that have been frequently studied and received generally consistent results. Beginning with the Big Five personality traits, anxious and avoidant attachment were both found to be negatively

Table 1 Meta-analytic results for anxious attachment

Variable	<i>k</i>	<i>N</i>	<i>r</i>	ρ	<i>SD_p</i>	95% CI		80% CR		<i>Q</i>
						LL	UL	LL	UL	
Agreeableness	11	9891	-.188	-.223	.043	-.264	-.185	-.287	-.166	102.309***
Conscientiousness	14	10,775	-.236	-.286	.102	-.352	-.223	-.426	-.149	35.219***
Extraversion	12	9971	-.150	-.184	.040	-.222	-.147	-.240	-.129	70.241***
Openness	12	10,104	-.087	-.105	.081	-.164	-.048	-.216	.005	49.200***
Neuroticism	14	10,524	.423	.491	.056	.452	.536	.418	.569	55.893***
Burnout	30	7415	.278	.318	.089	.276	.361	.201	.435	94.171***
Job satisfaction	26	4901	-.083	-.095	.147	-.164	-.027	-.288	.097	64.693***
Job stress	5	908	.342	.383	.035	.279	.487	.330	.436	9.744
Job performance	17	3923	-.094	-.110	.140	-.192	-.028	-.298	.078	82.299***
OCB	18	4468	-.088	-.100	.120	-.170	-.031	-.260	.060	27.824***
Self-efficacy	7	1650	-.236	-.272	0	-.330	-.218	-.274	-.274	53.273***
Trust in supervisor	5	2882	-.154	-.175	.112	-.326	-.023	-.347	-.002	5.163
LMX	8	1600	-.181	-.202	.056	-.282	-.122	-.282	-.123	4.305
Org commitment	14	2265	-.043	-.050	.229	-.193	.092	-.359	.258	23.219*
Turnover intentions	13	1732	.138	.171	.146	.062	.281	-.026	0.369	10.892
Work engagement	7	1563	-.125	-.142	.058	-.231	-.054	-.026	.369	43.099***

CR credibility interval, CI confidence interval. * $p < .05$, ** $p < .01$, *** $p < .001$

Table 2 Meta-analytic results for avoidant attachment

Variable	<i>k</i>	<i>N</i>	<i>r</i>	ρ	<i>SD_p</i>	95% CI		80% CR		<i>Q</i>
						LL	UL	LL	UL	
Agreeableness	11	9891	-.233	-.256	.089	-.342	-.209	-.399	-.153	176.190***
Conscientiousness	12	10,562	-.113	-.137	.0054	-.178	-.095	-.210	-.063	56.856***
Extraversion	12	9971	-.229	-.279	.069	-.331	-.228	-.374	-.185	34.690**
Openness	11	9891	-.102	-.122	.047	-.164	.240	-.404	.464	4.656
Neuroticism	13	10,311	.158	.183	.069	.134	.232	.088	.278	48.855***
Burnout	26	6883	.220	.252	.115	.198	.306	.100	.403	147.394***
Job satisfaction	24	4761	-.076	-.087	.209	-.182	.008	-.363	.190	211.295***
Job stress	5	908	.202	.226	.194	-.035	.487	-.072	.524	63.718***
Job performance	14	3523	-.112	-.131	.094	-.199	-.062	-.257	-.004	99.505***
OCB	16	4188	.158	-.105	.252	-.244	.034	-.443	.233	72.265***
Self-efficacy	7	1650	-.179	-.207	.125	-.341	-.073	-.386	-.028	24.404**
Trust in supervisor	7	3516	-.167	-.189	.163	-.347	-.031	-.424	.046	27.831***
LMX	6	1320	-.168	-.188	.226	-.438	.061	-.522	.146	23.666**
Org commitment	12	1985	.026	.030	.319	-.180	.240	-.404	.464	68.250***
Turnover intentions	13	1850	.091	.122	.199	-.024	.249	-.158	.383	52.515***
Work engagement	7	1761	-.143	-.162	.217	-.374	.049	-.475	.151	29.677***

CR credibility interval, CI confidence interval. * $p < .05$. ** $p < .01$. *** $p < .001$

Table 3 Meta-analytic results for secure attachment

Variable	<i>k</i>	<i>N</i>	<i>r</i>	ρ	<i>SD_p</i>	95% CI		80% CR		<i>Q</i>
						LL	UL	LL	UL	
Agreeableness	5	914	.226	.298	0	.203	.393	.298	.298	63.017***
Conscientiousness	7	1458	.159	.214	.036	.122	.306	.162	.265	47.240***
Extraversion	5	914	.351	.477	.055	.340	.614	.393	.561	27.578***
Openness	7	1529	.139	.186	.083	.072	.299	.066	.306	11.379*
Neuroticism	6	1127	-.295	-.381	.091	-.518	-.245	-.517	-.246	8.743*
Burnout	9	2617	-.238	-.303	.143	-.427	-.179	-.502	-.103	27.349***
Job satisfaction	8	1484	.127	.162	.262	-.071	.394	-.209	.532	88.200***
Job performance	6	1424	.174	.226	.173	.024	.427	-.029	.481	0.218
Turnover intentions	6	794	-.128	-.177	.344	-.559	.206	-.684	.331	11.461

CR credibility interval, CI confidence interval. * $p < .05$. ** $p < .01$. *** $p < .001$

related to extraversion (An. $r_c = -0.18$; Av. $r_c = -0.28$), openness to experience (An. $r_c = -0.11$; Av. $r_c = -0.12$), agreeableness (An. $r_c = -0.23$; Av. $r_c = -0.28$), and conscientiousness (An. $r_c = -0.29$; Av. $r_c = -0.14$) and positively to neuroticism (An. $r_c = 0.49$; Av. $r_c = 0.18$). These results were also confirmed with relationships with secure attachment that were in the opposite direction. Secure attachment was positively related to extraversion ($r_c = 0.48$), openness to experience ($r_c = 0.19$), agreeableness ($r_c = 0.30$), and conscientiousness ($r_c = 0.21$) and negatively related to neuroticism ($r_c = -0.38$).

Anxious and avoidant attachment were both positively related to burnout (An. $r_c = 0.32$; Av. $r_c = 0.25$). Anxious attachment was also positively related to job stress ($r_c = 0.38$), though the confidence interval for avoidant attachment contained zero for job stress ($r_c = 0.23$). Results with secure attachment confirmed the results for burnout,

showing a negative relationship as expected ($r_c = -0.30$). Anxious and avoidant attachment both showed negative relationships with job performance (An. $r_c = -0.11$; Av. $r_c = -0.13$), self-efficacy (An. $r_c = -0.27$; Av. $r_c = -0.21$), and trust in supervisor (An. $r_c = -0.17$; Av. $r_c = -0.19$). The results for job performance ($r_c = 0.23$) were further supported through positive relationships with secure attachment. Additionally, anxious attachment was negatively related to job satisfaction ($r_c = -0.10$), OCB ($r_c = -0.10$), and LMX ($r_c = -0.20$), though the confidence intervals for avoidant (JS: $r_c = -0.09$; OCB: $r_c = -0.10$; LMX: $r_c = -0.19$) and secure (JS: $r_c = 0.16$) attachment contained zero for these relationships.

Next, we turn to relationships with inconsistent findings in the literature. Neither anxious ($r_c = -0.05$) nor avoidant ($r_c = 0.03$) attachment were significantly related to organizational commitment. Additionally, only anxious attachment

was significantly related to turnover intentions ($r_c = 0.17$) and work engagement ($r_c = -0.14$). The confidence intervals contained zero for secure attachment with turnover intentions ($r_c = -0.18$) and for avoidant attachment with turnover intentions ($r_c = 0.11$) and work engagement ($r_c = -0.16$).

Aim 2

The second aim of this meta-analysis was to examine the incremental validity of attachment styles over the Big Five personality traits (Goldberg, 1990) in predicting job satisfaction, job performance, organizational commitment, and LMX. The dominance analysis results are reported in Table 4. First, for job satisfaction, the overall R^2 statistic was 0.606, suggesting that roughly 61% of the variance in job satisfaction was accounted for by the full model with

the eight predictors. The standardized dominance statistics then describe what percentage of the overall R^2 is attributable to each predictor. The predictor that accounted for the largest proportion of the overall fit statistic was neuroticism (38.5%), followed by openness (10.7%), anxious attachment (10.6%), secure attachment (10.1%), avoidant attachment (8.9%), conscientiousness (8.7%), extraversion (7.7%), and finally agreeableness (4.8%).

For job performance, the overall R^2 for the eight-predictor model was 0.601, suggesting that roughly 60% of the variance in job performance was accounted for by the tested model. The predictor that accounted for the greatest proportion of the overall R^2 was avoidant attachment (28.9%), followed by anxious attachment (19.5%), agreeableness (13.7%), neuroticism (11.0%), extraversion (7.6%),

Table 4 Dominance analyses results for job satisfaction, job performance, and organizational commitment

	Outcome	Overall fit statistic	Rank	General dominance	Standardized dominance
Job satisfaction	Agreeableness	0.606	8	0.029	0.048
	Conscientiousness		6	0.053	0.087
	Extraversion		7	0.047	0.077
	Neuroticism		1	0.233	0.385
	Openness		2	0.065	0.107
	Anxious Attachment		3	0.064	0.106
	Avoidant Attachment		5	0.054	0.089
	Secure attachment		4	0.061	0.101
Job performance	Agreeableness	0.601	3	0.082	0.137
	Conscientiousness		8	0.034	0.057
	Extraversion		5	0.046	0.076
	Neuroticism		4	0.066	0.110
	Openness		7	0.037	0.062
	Anxious Attachment		2	0.117	0.195
	Avoidant Attachment		1	0.174	0.289
	Secure attachment		6	0.044	0.074
Org commitment	Agreeableness	0.641	7	0.026	0.040
	Conscientiousness		5	0.041	0.064
	Extraversion		4	0.055	0.086
	Neuroticism		2	0.160	0.249
	Openness		8	0.016	0.024
	Anxious Attachment		3	0.067	0.107
	Avoidant Attachment		6	0.034	0.053
	Secure attachment		1	0.241	0.377
Leader-member exchange	Agreeableness	.777	8	.038	.049
	Conscientiousness		7	.041	.053
	Extraversion		6	.064	.082
	Neuroticism		5	.095	.122
	Openness		4	.100	.129
	Anxious Attachment		1	.157	.201
	Avoidant Attachment		2	.142	.182
	Secure attachment		3	.140	.180

secure attachment (7.4%), openness (6.2%), and conscientiousness (5.7%).

For organizational commitment, the overall R^2 for the eight-predictor model was 0.641, suggesting that roughly 64% of the variance in organizational commitment was accounted for by the tested model. The predictor that accounted for the largest proportion of the overall R^2 was secure attachment (37.7%), followed by neuroticism (24.9%), anxious attachment (10.7%), extraversion (8.6%), conscientiousness (6.4%), and avoidant attachment (5.3%), agreeableness (4.0%), and openness (2.4%).

Finally, for LMX the overall R^2 for the eight-predictor model was 0.777, meaning that roughly 78% of the variance in LMX was accounted for by the tested model. The predictor that accounted for the largest proportion of the overall R^2 was anxious attachment (20.1%), followed by avoidant attachment (18.2%), secure attachment (18.0%), openness (12.9%), neuroticism (12.2%), extraversion (8.2%), conscientiousness (5.3%), and agreeableness (4.9%).

Aim 3

The third aim of this meta-analysis was to examine whether attachment style impacts job performance, job satisfaction, organizational commitment, turnover intentions, and LMX through its impact on an employee's trust in their supervisor. Altogether, ten models were run, as each path model was run separately for anxious and avoidant attachment. For the indirect effect of anxious attachment on job performance, anxious attachment had both an indirect ($r_c = -0.027$; 95% CI [-0.154, -0.066]) and direct ($r_c = -0.083$) effect on job performance. The indirect path indicates that anxious attachment had a negative effect on trust in supervisor ($r_c = -0.175$), and trust in supervisor had a positive relationship with job performance ($r_c = 0.156$). Overall, anxious attachment had a significant total effect on job performance ($r_c = -0.110$). Similarly, avoidant attachment also had an indirect ($r_c = -0.028$; 95% CI [-0.174, -0.086]) and direct ($r_c = -0.102$) effect on job performance. Avoidant attachment was negatively related to trust in supervisor ($r_c = -0.189$), and trust in supervisor was positively related to job performance ($r_c = 0.151$). Overall, avoidant attachment had a negative total effect on job performance ($r_c = -0.131$).

With job satisfaction, anxious attachment had a significant indirect relationship through trust in supervisor ($r_c = -0.114$; 95% CI [-0.140, -0.051]), but did not have a significant direct effect ($r_c = 0.019$). Anxious attachment was negatively related to trust in supervisor ($r_c = -0.175$), which was then positively related to job satisfaction ($r_c = 0.653$). Overall, anxious attachment had a negative relationship with job satisfaction ($r_c = -0.095$).

For avoidant attachment, both the direct ($r_c = 0.037$) and indirect ($r_c = -0.124$; 95% CI [-0.131, -0.042]) effects on job satisfaction were significant. Interestingly, the direct effect was positive, while the indirect effect through trust in supervisor was negative. Avoidant attachment was negatively related to trust in supervisor ($r_c = -0.189$), and trust in supervisor was positively related to job satisfaction ($r_c = 0.657$). Overall, the total effect of avoidant attachment on job satisfaction was negative ($r_c = -0.087$).

Next, organizational commitment was examined with anxious attachment. Anxious attachment showed a positive main effect on organizational commitment ($r_c = 0.054$), but a negative indirect effect ($r_c = -0.105$; 95% CI [-0.095, -0.006]). Anxious attachment was negatively related to trust in supervisor ($r_c = -0.175$), and trust in supervisor was then positively related to organizational commitment ($r_c = 0.599$). Overall, the total effect of anxious attachment on organizational commitment was negative ($r_c = -0.050$). Interestingly, avoidant attachment showed a larger positive direct effect ($r_c = 0.147$) than negative indirect effect ($r_c = -0.117$; 95% CI [-0.014, 0.075]) on organizational commitment. The indirect path showed similar patterns to anxious attachment, with avoidant attachment being negatively related to trust in supervisor ($r_c = -0.189$), and trust in supervisor being positively related to organizational commitment ($r_c = 0.618$). However, the opposing indirect and direct effects led to an insignificant total effect ($r_c = 0.030$).

Next, turnover intentions showed positive direct effect ($r_c = 0.092$) and indirect effects ($r_c = 0.079$; 95% CI [0.128, 0.215]) of anxious attachment. Anxious attachment was negatively related to trust in supervisor ($r_c = -0.175$), and trust in supervisor was negatively related to turnover intentions ($r_c = -0.454$). The total effect of anxious attachment on turnover intentions was positive ($r_c = 0.171$). Avoidant attachment did not have a significant direct effect on turnover intentions ($r_c = 0.024$), but did have a significant indirect effect ($r_c = 0.088$; 95% CI [0.069, 0.157]). Avoidant attachment was negatively related to trust in supervisor ($r_c = -0.189$), and trust in supervisor was negatively related to turnover intentions ($r_c = -0.465$). The total effect of avoidant attachment on turnover intentions was positive ($r_c = 0.112$).

Lastly, anxious attachment showed a negative indirect ($r_c = -0.132$; 95% CI [-0.245, -0.158]) and direct ($r_c = -0.070$) effect on LMX. Anxious attachment reduced trust in supervisor ($r_c = -0.175$), and trust in supervisor was positively related to LMX ($r_c = 0.758$). Together, the indirect and direct effect led to a negative total effect of anxious attachment on LMX ($r_c = -0.202$). Similarly, avoidant attachment had a negative direct ($r_c = -0.044$) and indirect ($r_c = -0.144$; 95% CI [-0.232, -0.144]) effect on LMX. Avoidant attachment

Table 5 Results for mediation structural equation model through trust in supervisor

Variable	<i>Anxious</i>						$R^{2'}$
	R^2	<i>a</i>	<i>b</i>	<i>ab</i>	<i>c'</i>	<i>c</i>	
Job satisfaction	.423	-.175***	.653***	-.114***	.019	-.095***	.030
Job performance	.036	-.175***	.156***	-.027***	-.083***	-.110***	.030
Org commitment	.351	-.175***	.599***	-.105***	.054**	-.050*	.030
Turnover intentions	.229	-.175***	-.454***	.079***	.092***	.171***	.030
LMX	.598	-.175***	.758***	-.132***	-.070***	-.202***	.030

Harmonic mean = 3401.178. *a* = first stage mediation effect; *b* = second stage mediation effect; *ab* = indirect effect; *c'* = direct effect; *c* = overall effect; $R^{2'}$ = R^2 for indirect effect

* $p < .05$, ** $p < .01$, *** $p < .001$

Table 6 Results for mediation structural equation model through trust in supervisor

Variable	<i>Avoidant</i>						$R^{2'}$
	R^2	<i>a</i>	<i>b</i>	<i>ab</i>	<i>c'</i>	<i>c</i>	
Job satisfaction	.424	-.189***	.657***	-.124***	.037*	-.087***	.036
Job performance	.039	-.189***	.151***	-.028***	-.102***	-.131***	.036
Org commitment	.369	-.189***	.618***	-.117***	.147***	.030	.036
Turnover intentions	.221	-.189***	-.465***	.088***	.024	.112***	.036
LMX	.595	-.189***	.762***	-.144***	-.044**	-.188***	.036

Harmonic mean = 3401.178. *a* = first stage mediation effect; *b* = second stage mediation effect; *ab* = indirect effect; *c'* = direct effect; *c* = overall effect; $R^{2'}$ = R^2 for indirect effect

* $p < .05$, ** $p < .01$, *** $p < .001$

reduced trust in supervisor ($r_c = -0.189$), and trust in supervisor was positively related to LMX ($r_c = 0.762$). The total effect of avoidant attachment on LMX was negative ($r_c = -0.188$). The results of the path models are shown for anxious attachment in Table 5 and avoidant attachment in Table 6.

Sensitivity Analyses

To examine outliers, we first created forest plots of all the tested relationships. Of these, 11 were flagged as potentially containing outliers. We then conducted leave-one-out analyses to further investigate these 11 relationships, but found nothing of concern regarding outliers. Thus, no samples were excluded.

To examine potential publication bias, we first created funnel plots for relationships with a *k* of greater than ten, as power is an issue for plots made with fewer than ten studies (Sterne et al., 2011). Of the 22 relationships examined using funnel plots, 6 were flagged as potentially asymmetrical. To follow up on the 6 potentially asymmetrical funnel plots, we then conducted moderation analyses for these relationships examining the differences between publication type (i.e., published, unpublished thesis, published dissertation). These analyses were all insignificant, suggesting that publication bias was unlikely.

Discussion

Summary of Results

Aim 1

The first aim of this meta-analysis was to confirm and clarify the effect sizes for the relationships between attachment style and various workplace correlates. The strongest support was found for all the Big Five personality traits, burnout, and job performance as correlates of attachment style. These correlates were significant for both anxious and avoidant attachment and were confirmed by relationships of the opposite valence with secure attachment. Of these correlates, anxious and avoidant attachment were positively related to neuroticism and burnout, and negatively related to conscientiousness, agreeableness, extraversion, openness, and job performance. These relationships, other than job performance, were also relatively strong compared to the effect sizes for other correlates.

There were several other relationships that received support, though were not significant for both anxious and avoidant or not bolstered by an opposite relationship with secure attachment, indicating mixed results for the impact of attachment style. Both anxious and avoidant attachment were negatively related to self-efficacy and trust in

supervisor. However, these relationships were either not able to be examined with secure attachment or were not significant for secure attachment. Additionally, there were several correlates that had significant relations with anxious attachment only. Anxious attachment was positively related to job stress and turnover intentions, and negatively related to job satisfaction, organizational citizenship behavior, LMX, and work engagement.

There were also several relationships that were not supported at the meta-analytic level. Neither anxious nor avoidant attachment were related to organizational commitment. Further, avoidant attachment was not related to job stress, job satisfaction, organizational citizenship behavior, LMX, turnover intention, or work engagement. Given these mixed and non-significant results, we urge caution when advocating for attachment styles as a panacea for understanding all organizational phenomena. The impact of attachment styles on the aforementioned work outcomes may depend on an interplay of attachment styles and various contextual factors, such as organizational culture or policies and job characteristics, that this study was not able to capture. While attachment theory offers valuable insights into understanding some of the work correlates we describe above, its relationship with other work-related variables may require a more nuanced examination of contextual factors that work with attachment style in shaping employees' job attitudes and motivation.

Aim 2

We also aimed to shed light on the utility of attachment style above and beyond the Big Five personality traits for several outcomes. The dominance analysis results showed that attachment style had the most impact on the overall R^2 for each of the tested models except for job satisfaction. Namely, avoidant attachment had the highest dominance statistic for job performance, secure attachment had the highest dominance statistic for organizational commitment, and anxious attachment had the highest dominance statistic for LMX. The influence of attachment styles was especially pronounced for LMX, for which attachment styles were the top three predictors. Given the nature of LMX as a relational construct and its dependence on trust (Graen & Uhl-Bien, 1995), it is understandable why attachment style would have such a large impact. Additionally, secure attachment accounted for a large proportion of the variance in organizational commitment (37.7%). Like LMX, organizational commitment has been proposed to have trust as one of its key antecedents (Nyhan, 1999); thus, the trusting component of secure attachment (Bowlby, 1969) is likely impactful for the development of organizational commitment.

Aim 3

Finally, we tested several meta-analytic path models to examine the mediating effect of trust in supervisor on some of the tested correlates. Overall, we found support for the mediating effect of trust in supervisor between both anxious and avoidant attachment and all the tested correlates (job performance, job satisfaction, turnover intentions, organizational commitment, and LMX). Interestingly, the direct effect was occasionally not significant, or was significant in the opposite direct (i.e., a positive direct effect while the indirect effect was negative). This suggests that it is important to consider the mechanisms through which attachment style impacts workplace outcomes, as capturing only the direct effect may not explain the full story.

Research and Practical Implications

The current meta-analysis makes several contributions to the literature and provides key practical implications. First, by conducting the first quantitative synthesis (to our knowledge) of the relationships between attachment styles and work correlates, we have established the most precise estimates of these relationships. The results support that, although often overlooked in organizational research, attachment styles can provide useful insight into workplace outcomes (Harms, 2011). This seems to be especially true of anxious attachment; however, as the bivariate correlations indicate, scholars and practitioners should use caution when describing the relations among attachment styles and their correlates.

The large effect sizes for anxious attachment with job stress and burnout provide support for the proposed job-stress mechanism through which insecure attachment impacts work outcomes. That is, anxious attachment in particular is related to difficulty managing stressors and regulating negative emotions (Gillath et al., 2005; Harms, 2011). This has been proposed to relate to increased job stress and burnout for those with insecure attachment styles (Collins, 1996; Richards & Schat, 2011). Our results support these relationships and highlight the importance for managers and leaders to acknowledge the impact of attachment styles on workplace variables, especially for roles that have a large interpersonal component. Notably, individuals with anxious attachment appear to struggle with managing stressors and regulating negative emotions. Therefore, organizational leaders should prioritize understanding employees' attachment styles when designing interventions to alleviate job stress and burnout. Implementing strategies aimed at enhancing coping mechanism at work may be especially beneficial for individuals with anxious attachment styles.

We further extended our findings by examining the incremental validity of attachment styles over the Big Five and

testing mediation path models. The results of the dominance analyses suggested that attachment styles were critical predictors of job performance, organizational commitment, and LMX; thus, it is possible that attachment style represents a unique individual difference that is not captured by other, commonly examined individual differences such as the Big Five. Specifically, the perceptions of self-worth and perceptions of others as good and reliable are not captured by other personality measures often used in organizational research and could thus offer unique insight into organizational behavior. These results support the warnings against viewing attachment style as a composite of personality (Fraley & Shaver, 2008; Harms, 2011) and align with previous findings that attachment styles hold predictive power over the Big Five traits (Nofle & Shaver, 2006; Roisman et al., 2007).

We also showed that trust in one's supervisor is one important mechanism through which attachment style impacts organizational outcomes. Previous research has confirmed the relationship between low trust in one's supervisor and organizational outcomes (e.g., Dirks & Ferrin, 2002), and here we connected the findings with attachment research to establish the mediating mechanism. This provides support for the relationships proposed by attachment theory and clearly links attachment theory to organizational theories. This is critical for highlighting how attachment is impactful at work, and why attachment styles should be included in our study of individual differences that impact workplace outcomes.

Lastly, given that trust in one's supervisor mediates the relationship between attachment style and important organizational outcomes, managers should consider employees' attachment styles and tailor their leadership approaches accordingly. Specifically, managers can adjust their communication styles, feedback mechanisms, and support strategies to accommodate the needs and preferences of their employees with different attachment styles. Especially for employees with insecure attachment, it is crucial for organizational leaders to foster trust and psychological safety in relationships with their employees to improve trust in their supervisor. For employees with an anxious attachment style, they would likely thrive with leadership that is consistent, transparent and honest, and understanding. These qualities of leadership would reduce ambiguity in their relationship with their leader and thus reduce their attachment-related anxiety. Similarly, those with an avoidant attachment style would likely thrive with leaders who are dependable, committed, and consistent. These qualities in a leader can prevent avoidant followers from feeling "let down" by their leader and prompting the urge to distance themselves and rely only on themselves. By establishing trust in the leader–follower relationship by considering the follower's unique attachment needs, workplace outcomes such as job performance and organizational commitment can be improved.

Limitations and Future Research Directions

There are a few limitations of the current research that should be addressed. First, the meta-analytic structural equation modeling technique we utilized here (Viswesvaran & Ones, 1995) has some notable limitations. Specifically, this method suffers from issues associated with unbalanced sample sizes of the coefficients included in the correlation matrix (Alamolhoda et al., 2017). Future research could replicate our findings using other methods that can alleviate this heterogeneity problem, such as two-stage structural equation modeling (Cheung & Chan, 2009) or full information meta-analytical structural equation modeling (Yu et al., 2018).

Second, the heterogeneity analyses resulted in many significant Q statistics, as well as several credibility intervals that included zero, for the tested relationships, which suggests that there is heterogeneity in the tested studies and that there may be moderators in the relationships that were not examined. However, the studies examined here did not present any potential moderators that we could examine at the meta-analytic level, as there was not enough overlap between third variables included in the primary studies. This is a limitation of the current meta-analysis, as we were not able to further investigate heterogeneity between studies, and many of the inconsistent findings thus remained unclarified. However, this is also a limitation of the literature. Our findings suggest that heterogeneity exists and that the relationships between attachment styles and workplace correlates may not be consistent across situations or between people. Thus, future research at the primary level should include theoretical moderators in their examinations of attachment at work to tease apart the true relationships between attachment styles and workplace behavior and other variables. For example, the insecure attachment styles are activated by inconsistent, ambiguous, or distant behavior from others (Keller, 2003). That is, when relationships appear to be stable, consistent, and responsive, the attachment system will not be activated (Bowlby, 1969). Thus, in work environments in which leaders and peers are consistently supportive, available, transparent, and genuine, the insecure attachment patterns may not be activated, and the impact of these attachment styles on work outcomes may not be as strong. Future research could examine the moderating effect of situational and relational factors on the activation and impact of attachment insecurity.

Furthermore, given the availability of primary studies, we were only able to test one potential mechanism through which attachment impacts workplace outcomes. Although trust in one's supervisor is perhaps the most theoretically supported mechanism, there are other possibilities that should be examined, and compared with trust in supervisor, in future research. For example, it is possible that job stress, burnout, or cognitive load could also act as a mediating

mechanism through which attachment impacts workplace outcomes. Those with insecure attachment styles are more likely to have relationship difficulties, less access to social support as a coping strategy for stressors, more fear of failure and rejection, and be overly engrossed in work (Hardy & Barkham, 1994; Harms, 2011; Hazan & Shaver, 1990; Vîrgă et al., 2019). These difficulties at work are likely to lead to increased stress and burnout (Collins, 1996; Harms, 2011; Richards & Schat, 2011), which can then impact performance (Vîrgă et al., 2019). Given the found discrepancies between the valence and significance of the direct effects compared to the mediating effects, it seems that exploring mediating mechanisms further will be needed to fully understand the impact of attachment styles on various workplace outcomes. Thus, future research should expand on the mediating mechanisms tested here to further illuminate how attachment style comes into play in a work setting.

Additionally, there were several construct-specific moderators that we were unable to examine with this meta-analysis that could be examined in future research. For example, burnout was well-represented in the literature, yet future research could examine the relationship between attachment styles and the three dimensions of burnout—emotional exhaustion, depersonalization, and reduced personal accomplishment (Maslach et al., 2001). It is possible that the insecure attachment styles relate differentially to these three dimensions and that these three dimensions mediate differentially to various other workplace outcomes. Thus, given the important role that burnout seems to play in the relationship between attachment styles and workplace outcomes, a deeper look at burnout and its dimensions could be useful. Similarly, differential relationships could exist between attachment style and the separate dimensions of job satisfaction (e.g., satisfaction with supervisor, coworkers, the job itself) and organizational commitment (affective, continuance, normative). The results of the mediation analyses supported the relationship between these constructs in their composite form and attachment styles through trust in supervisor. Thus, future research could expand these results by breaking job satisfaction and organizational commitment into their separate components when examining trust as a mediator, as well as when examining other potential mediators as mentioned above with these separate dimensions.

Conclusion

The current meta-analysis sought to synthesize research across various disciplines regarding the relationship between attachment styles and work-related outcomes and personality traits. Additionally, we found support for the importance of attachment style above and beyond personality in predicting job performance, job satisfaction, and organizational

commitment. We also found support for trust in one's supervisor as a mediator between attachment style and various workplace outcomes. Overall, the results suggest that attachment style can be a useful predictor of various important work correlates and outcomes.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10869-024-09960-9>.

Declarations

Conflict of Interest The authors declare no competing interests.

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References

(References that were included in the meta-analysis can be found in the supplemental materials)

- Ainsworth, M. D. S., Blehar, M. C., Waters, E., & Wall, S. (1978). *Patterns of attachment: A psychological study of the strange situation*. Lawrence Erlbaum.
- Alamolhoda, M., Ayatollahi, S. M. T., & Bagheri, Z. (2017). A comparative study of the impacts of unbalanced sample sizes on the four synthesized methods of meta-analytic structural equation modeling. *BMC Research Notes*, 10(1), 1–12. <https://doi.org/10.1186/s13104-017-2768-5>
- Allen, T. D., Shockley, K. M., & Poteat, L. (2010). Protégé anxiety attachment and feedback in mentoring relationships. *Journal of Vocational Behavior*, 77(1), 73–80. <https://doi.org/10.1016/j.jvb.2010.02.007>
- Banerjee-Batist, R., & Reio, T. G. (2016). Attachment and mentoring: Relations with junior faculty's organizational commitment and intent to turnover. *Journal of Management Development*, 35(3), 360–381. <https://doi.org/10.1108/JMD-02-2015-0015>
- Barrick, M. R., & Mount, M. K. (1991). The big five personality dimensions and job performance: A meta-analysis. *Personnel Psychology*, 44(1), 1–26. <https://doi.org/10.1111/j.1744-6570.1991.tb00688.x>
- Bartholomew, K. (1997). Adult attachment processes: Individual and couple perspectives. *British Journal of Medical Psychology*, 70(3), 249–263. <https://doi.org/10.1111/j.2044-8341.1997.tb01903.x>
- Berlanda, S., Pedrazza, M., Trifiletti, E., & Fraizzoli, M. (2019). Sources of physicians' well-being: An explorative qualitative study. *TPM - Testing, Psychometrics, Methodology in Applied Psychology*, 25, 121–137. <https://doi.org/10.4473/TPM25.1.7>

- Berson, Y., Dan, O., & Yammarino, F. J. (2006). Attachment style and individual differences in leadership perceptions and emergence. *The Journal of Social Psychology, 146*(2), 165–182. <https://doi.org/10.3200/SOCP.146.2.165-182>
- Blau, P. M. (1964). *Exchange and power in social life*. Wiley.
- Boatwright, K. J., Lopez, F. G., Sauer, E. M., VanDerWege, A., & Huber, D. M. (2010). The influence of adult attachment styles on workers' preferences for relational leadership behaviors. *The Psychologist-Manager Journal, 13*(1), 1–14. <https://doi.org/10.1080/10887150903316271>
- Bowlby, J. (1969). *Attachment and Loss, Vol. 1: Attachment*. New York, NY: Basic Books.
- Byrne, Z., Albert, L., Manning, S., & Desir, R. (2017). Relational models and engagement: An attachment theory perspective. *Journal of Managerial Psychology, 32*(1), 30–44. <https://doi.org/10.1108/JMP-01-2016-0006>
- Cheung, M.W.-L., & Chan, W. (2009). A two-stage approach to synthesizing covariance matrices in meta-analytic structural equation modeling. *Structural Equation Modeling: A Multidisciplinary Journal, 16*(1), 28–53. <https://doi.org/10.1080/10705510802561295>
- Choi, D., Oh, I.-S., & Colbert, A. E. (2015). Understanding organizational commitment: A meta analytic examination of the roles of the five-factor model of personality and culture. *Journal of Applied Psychology, 100*(5), 1542–1567. <https://doi.org/10.1037/apl0000014>
- Chopik, W. J. (2015). Relational attachment and ethical workplace decisions: The mediating role of emotional burnout. *Personality and Individual Differences, 75*, 160–164. <https://doi.org/10.1016/j.paid.2014.11.007>
- Collins, N. L. (1996). Working models of attachment: Implications for explanation, emotion, and behavior. *Journal of Personality and Social Psychology, 71*(4), 810–832.
- Dahling, J. J., & Librizzi, U. A. (2015). Integrating the theory of work adjustment and attachment theory to predict job turnover intentions. *Journal of Career Development, 42*(3), 215–228. <https://doi.org/10.1177/0894845314545169>
- Dahlke, J. A., & Wiernik, B. M. (2019). Psychmeta: An R package for psychometric meta-analysis. *Applied Psychological Measurement, 43*(5), 415–416. <https://doi.org/10.1177/0146621618795933>
- Davidovitz, R., Mikulincer, M., Shaver, P. R., Izsak, R., & Popper, M. (2007). Leaders as attachment figures: Leaders' attachment orientations predict leadership-related mental representations and followers' performance and mental health. *Journal of Personality and Social Psychology, 93*(4), 632–650. <https://doi.org/10.1037/0022-3514.93.4.632>
- Desivilya, H. S., Sabag, Y., & Ashton, E. (2006). Prosocial tendencies in organizations: The role of attachment styles and organizational justice in shaping organizational citizenship behavior. *International Journal of Organizational Analysis, 14*(1), 22–42. <https://doi.org/10.1108/10553180610739731>
- Dirks, K. T., & Ferrin, D. L. (2002). Trust in leadership: Meta-analytic findings and implications for research and practice. *Journal of Applied Psychology, 87*(4), 611–628. <https://doi.org/10.1037/0021-9010.87.4.611>
- Doverspike, D., Hollis, L., Justice, A., & Polomsky, M. (1997). Correlations between leadership styles as measured by the least preferred co-worker scale and adults' attachment styles. *Psychological Reports, 81*, 1148–1150.
- Dozier, M., & Kobak, R. R. (1992). Psychophysiology in attachment interviews: Converging evidence for deactivating strategies. *Child Development, 63*, 1473–1480.
- Dulebohn, J. H., Bommer, W. H., Liden, R. C., Brouer, R. L., & Ferris, G. R. (2012). A meta analysis of antecedents and consequences of leader-member exchange: Integrating the past with an eye toward the future. *Journal of Management, 38*(6), 1715–1759. <https://doi.org/10.1177/0149206311415280>
- Emerson, R. M. (1976). Social exchange theory. *Annual Review of Sociology, 2*, 335–362. <https://doi.org/10.1146/annurev.so.02.080176.002003>
- Falvo, R., Colledani, D., & Capozza, D. (2021). Denying full humanity to patients and nurses' well-being: The moderating role of attachment security. *TPM - Testing, Psychometrics, Methodology in Applied Psychology, 28*(3), 329–342. <https://doi.org/10.4473/TPM28.3.4>
- Feeney, J., & Noller, P. (1996). *Adult attachment*. Thousand Oaks, CA: Sage Publications. <https://doi.org/10.4135/9781452243276>
- Finkel, E. J., & Simpson, J. A. (2015). Editorial overview: Relationship science. *Current Opinion in Psychology, 1*, 5–9. <https://doi.org/10.1016/j.copsyc.2015.01.012>
- Fraley, R. C., & Shaver, P. R. (2008). Attachment theory and its place in contemporary personality research. In O. John & R. W. Robins (Eds.), *Handbook of Personality: Theory and Research* (3rd ed., pp. 518–541). Guilford Press.
- Frazier, M. L., Gooty, J., Little, L. M., & Nelson, D. L. (2015). Employee attachment: Implications for supervisor trustworthiness and trust. *Journal of Business and Psychology, 30*(2), 373–386. <https://doi.org/10.1007/s10869-014-9367-4>
- Furness, M. P. (2020). What predicts career adaptability?: An application of achievement goal theory and adult attachment theory. *Journal of Career Development, 47*(6), 671–685. <https://doi.org/10.1177/0894845318815610>
- Geller, D., & Bamberger, P. (2009). Bringing avoidance and anxiety to the job: Attachment style and instrumental helping behavior among co-workers. *Human Relations, 62*(12), 1803–1827. <https://doi.org/10.1177/0018726709337524>
- Gillath, O., Bunge, S. A., Shaver, P. R., Wendelken, C., & Mikulincer, M. (2005). Attachment-style differences in the ability to suppress negative thoughts: Exploring the neural correlates. *NeuroImage, 28*(4), 835–847. <https://doi.org/10.1016/j.neuroimage.2005.06.048>
- Goldberg, L. R. (1990). An alternative “description of personality”: The big-five factor structure. *Journal of Personality and Social Psychology, 59*(6), 1216–1229.
- Graen, G. B., & Uhl-Bien, M. (1995). Relationship-based approach to leadership: Development of leader-member exchange (LMX) theory of leadership over 25 years: Applying a multi level multi-domain perspective. *The Leadership Quarterly, 6*(2), 219–247. [https://doi.org/10.1016/1048-9843\(95\)90036-5](https://doi.org/10.1016/1048-9843(95)90036-5)
- Gruda, D., & Kafetsios, K. (2019). Attachment orientations guide the transfer of leadership judgments: Culture matters. *Personality and Social Psychology Bulletin, 46*(4), 525–546. <https://doi.org/10.1177/0146167219865514>
- Hardy, G. E., & Barkham, M. (1994). The relationship between interpersonal attachment styles and work difficulties. *Human Relations, 47*(3), 263–281.
- Harms, P. D. (2011). Adult attachment styles in the workplace. *Human Resource Management Review, 21*(4), 285–296. <https://doi.org/10.1016/j.hrmr.2010.10.006>
- Haroush, O. B., & Koslowsky, M. (2020). Attachment style as a predictor of burnout and work engagement among health professional caregivers. *Open Access Journal of Biogenetic Science and Research, 5*(4), 1–9. <https://doi.org/10.46718/JBGSR.2020.05.000127>
- Hazan, C., & Shaver, P. (1987). Romantic love conceptualized as an attachment process. *Journal of Personality and Social Psychology, 52*, 511–524.

- Hazan, C., & Shaver, P. R. (1990). Love and work: An attachment-theoretical perspective. *Journal of Personality and Social Psychology*, 59(2), 270–280.
- Hazan, C., & Shaver, P. R. (1994). Attachment as an organizational framework for research on close relationships. *Psychological Inquiry*, 5(1), 1–22. https://doi.org/10.1207/s15327965pli0501_1
- Hiebler-Ragger, M., Nausner, L., Blaha, A., Grimmer, K., Korlath, S., Mernyi, M., & Unterrainer, H. F. (2020). The supervisory relationship from an attachment perspective: Connections to burnout and sense of coherence in health professionals. *Clinical Psychology & Psychotherapy*, 28(1), 124–136. <https://doi.org/10.1002/cpp.2494>
- Hudson, D. L. (2013). Attachment theory and leader-follower relationships. *The Psychologist-Manager Journal*, 16(3), 147–159. <https://doi.org/10.1037/mgr0000003>
- Huedo-Medina, T. B., Sánchez-Meca, J., Marín-Martínez, F., & Botella, J. (2006). Assessing heterogeneity in meta-analysis: Q statistic or I² index? *Psychological Methods*, 11(2), 193–206. <https://doi.org/10.1037/1082-989X.11.2.193>
- Hunter, J. E., & Schmidt, F. L. (2004). *Methods of meta-analysis: Correcting error and bias in research findings* (2nd ed). Thousand Oaks, CA: Sage.
- Jackson, E. M. (2008). *Integrating leader-member exchange and organizational justice: Why justice depends on relationship quality*. [Unpublished Thesis]. University of South Florida.
- Jang, H.-R., Kim, K.-S., Ahn, H.-M., & Jhun, K.-C. (2013). The influence of culinary employee's interpersonal attachment on their job satisfaction and job attachment. *The Korean Journal of Culinary Research*, 19(1), 1–11.
- Johnston, M. A. (2000). Delegation and organizational structure in small businesses: Influences of manager's attachment patterns. *Group & Organization Management*, 25(1), 4–21. <https://doi.org/10.1177/1059601100251002>
- Johnstone, M., & Feeney, J. A. (2015). Individual differences in responses to workplace stress: The contribution of attachment theory: The contribution of attachment theory. *Journal of Applied Social Psychology*, 45(7), 412–424. <https://doi.org/10.1111/jasp.12308>
- Judge, T. A., Heller, D., & Mount, M. K. (2002). Five-factor model of personality and job satisfaction: A meta-analysis. *Journal of Applied Psychology*, 87(3), 530–541. <https://doi.org/10.1037/0021-9010.87.3.530>
- Kafetsios, K., Athanasiadou, M., & Dimou, N. (2014). Leaders' and subordinates' attachment orientations, emotion regulation capabilities and affect at work: A multilevel analysis. *The Leadership Quarterly*, 25(3), 512–527. <https://doi.org/10.1016/j.leaqua.2013.11.010>
- Kale, E. (2020). Attachment styles and job performance in the hospital industry: The mediating role of general self-efficacy. *Journal of Human Resources in Hospitality & Tourism*, 19(1), 23–42. <https://doi.org/10.1080/15332845.2020.1672244>
- Keller, T. (2003). Parental images as a guide to leadership sensemaking: An attachment perspective on implicit leadership theories. *The Leadership Quarterly*, 14(2), 141–160. [https://doi.org/10.1016/S1048-9843\(03\)00007-9](https://doi.org/10.1016/S1048-9843(03)00007-9)
- Lavy, S. (2014). supervisor security provision: Correlates and related mechanisms. *Psychological Reports*, 114(3), 758–783. <https://doi.org/10.2466/01.21.PR0.114k30w8>
- Littman-Ovadia, H., Oren, L., & Lavy, S. (2013). Attachment and autonomy in the workplace: New insights. *Journal of Career Assessment*, 21(4), 502–518. <https://doi.org/10.1177/1069072712475282>
- Lopez, F. G., & Ramos, K. (2016). Predicting well-being in managers: Test of a positive psychology model. *The Journal of Positive Psychology*, 11(4), 378–387. <https://doi.org/10.1080/17439760.2015.1092571>
- Luchman, J. (2023). domir: Tools to support relative importance analysis, R package version 1.1.1. Retrieved December 10, 2023 from <https://CRAN.R-project.org/package=domira>
- Luke, M. A., Carnelley, K. B., & Sedikides, C. (2020). Attachments in the workplace: How attachment security in the workplace benefits the organisation. *European Journal of Social Psychology*, 50(5), 1046–1064. <https://doi.org/10.1002/ejsp.2652>
- Maslach, C., Schaufeli, W. B., & Leiter, M. P. (2001). Job burnout. *Annual Review of Psychology*, 52, 397–422.
- Mayer, R. C., & Gavin, M. B. (2005). Trust in management and performance: Who minds the shop while the employees watch the boss? *Academy of Management Journal*, 48(5), 874–888. <https://doi.org/10.5465/amj.2005.18803928>
- Mayseless, O., & Popper, M. (2007). Reliance on leaders and social institutions: An attachment perspective. *Attachment & Human Development*, 9(1), 73–93. <https://doi.org/10.1080/14616730601151466>
- Mikulincer, M., & Shaver, P. R. (2003). the attachment behavioral system in adulthood: activation, psychodynamics, and interpersonal processes. In *Advances in Experimental Social Psychology* (Vol. 35, pp. 53–152). Elsevier. [https://doi.org/10.1016/S00652601\(03\)01002-5](https://doi.org/10.1016/S00652601(03)01002-5)
- Mikulincer, M., & Florian, V. (1995). Appraisal of and coping with a real-life stressful situation: The contribution of attachment styles. *Personality and Social Psychology Bulletin*, 21(4), 406–414.
- Mikulincer, M., & Shaver, P. R. (2007). *Attachment in adulthood: Structure, dynamics, and change*. Guilford Press.
- Mikulincer, M., & Shaver, P. R. (2016). *Attachment in adulthood: Structure, dynamics, and change (Second edition)*. Guilford Press.
- Nae, E. Y., & Choi, B. K. (2022). Career satisfaction, subjective well-being and turnover intention: An attachment style perspective. *Baltic Journal of Management*, 17(1), 1–17. <https://doi.org/10.1108/BJM-03-2021-0084>
- Ng, T. W. H. (2015). The incremental validity of organizational commitment, organizational trust, and organizational identification. *Journal of Vocational Behavior*, 88, 154–163. <https://doi.org/10.1016/j.jvb.2015.03.003>
- Noftle, E. E., & Shaver, P. R. (2006). Attachment dimensions and the big five personality traits: Associations and comparative ability to predict relationship quality. *Journal of Research in Personality*, 40(2), 179–208. <https://doi.org/10.1016/j.jrp.2004.11.003>
- Nyhan, R. C. (1999). Increasing affective organizational commitment in public organizations: The key role of interpersonal trust. *Review of Public Personnel Administration*, 19(3), 58–70. <https://doi.org/10.1177/0734371X9901900305>
- Oh, I.-S. (2020). Beyond meta-analysis: Secondary uses of meta-analytic data. *Annual Review of Organizational Psychology and Organizational Behavior*, 7, 125–153.
- Pines, A. M. (2004). Adult attachment styles and their relationship to burnout: A preliminary, cross-cultural investigation. *Work & Stress*, 18(1), 66–80. <https://doi.org/10.1080/02678370310001645025>
- Popper, M., & Mayseless, O. (2003). Back to basics: Applying a parenting perspective to transformational leadership. *The Leadership Quarterly*, 14(1), 41–65. [https://doi.org/10.1016/S1048-9843\(02\)00183-2](https://doi.org/10.1016/S1048-9843(02)00183-2)
- Raby, K. L., Fraley, R. C., & Roisman, G. I. (2021). Categorical or dimensional measures of attachment?: Insights from factor-analytic and taxometric research. In R. A. Thompson, J. A. Simpson, & L. J. Berlin (Eds.), *Attachment: The fundamental questions* (pp. 70–77). The Guilford Press.
- Reizer, A. (2015). influence of employees' attachment styles on their life satisfaction as mediated by job satisfaction and burnout. *The*

- Journal of Psychology*, 149(4), 356–377. <https://doi.org/10.1080/00223980.2014.881312>
- Reizer, A. (2019). bringing self-kindness into the workplace: Exploring the mediating role of self-compassion in the associations between attachment and organizational outcomes. *Frontiers in Psychology*, 10, 1148. <https://doi.org/10.3389/fpsyg.2019.01148>
- Rich, G. A. (1997). the sales manager as a role model: Effects on trust, job satisfaction, and performance of salespeople. *Journal of the Academy of Marketing Science*, 25(4), 319–328. <https://doi.org/10.1177/0092070397254004>
- Richards, D. A., & Hackett, R. D. (2012). Attachment and emotion regulation: Compensatory interactions and leader–member exchange. *The Leadership Quarterly*, 23(4), 686–701. <https://doi.org/10.1016/j.leaqua.2012.03.005>
- Richards, D. A., & Schat, A. C. H. (2011). Attachment at (not to) work: Applying attachment theory to explain individual behavior in organizations. *Journal of Applied Psychology*, 96(1), 169–182. <https://doi.org/10.1037/a0020372>
- Ricketta, M. (2008). The causal relation between job attitudes and performance: A meta-analysis of panel studies. *Journal of Applied Psychology*, 93(2), 472–481. <https://doi.org/10.1037/0021-9010.93.2.472>
- Rivera, P. C. (2020). *The relationship between subtle sexism and women's careers explained by cognitive processes and moderated by attachment styles*. California State University.
- Roisman, G. I., Holland, A., Fortuna, K., Fraley, R. C., Clausell, E., & Clarke, A. (2007). The adult attachment interview and self-reports of attachment style: An empirical rapprochement. *Journal of Personality and Social Psychology*, 92(4), 678–697. <https://doi.org/10.1037/0022-3514.92.4.678>
- Ronen, S., & Mikulincer, M. (2009). Attachment orientations and job burnout: The mediating roles of team cohesion and organizational fairness. *Journal of Social and Personal Relationships*, 26(4), 549–567. <https://doi.org/10.1177/0265407509347930>
- Rosseel, Y. (2012). lavaan: An R package for structural equation modeling. *Journal of Statistical Software*, 48(2), 1–36. <https://doi.org/10.18637/jss.v048.i02>
- Schirmer, L. L., & Lopez, F. G. (2001). Probing the social support and work strain relationship among adult workers: Contributions of adult attachment orientations. *Journal of Vocational Behavior*, 59(1), 17–33. <https://doi.org/10.1006/jvbe.2000.1777>
- Schusterschitz, C., Geser, W., Nöhammer, E., & Stummer, H. (2011). Securely attached, strongly committed? On the influence of attachment orientations on organizational commitment. *German Journal of Human Resource Management: Zeitschrift Für Personalforschung*, 25(4), 335–355. <https://doi.org/10.1177/239700221102500407>
- Scrima, F., Di Stefano, G., Guarnaccia, C., & Lorito, L. (2015). The impact of adult attachment style on organizational commitment and adult attachment in the workplace. *Personality and Individual Differences*, 86, 432–437. <https://doi.org/10.1016/j.paid.2015.07.013>
- Shaver, P. R., & Brennan, K. A. (1992). Attachment styles and the “big five” personality traits: Their connections with each other and with romantic relationship outcomes. *Personality and Social Psychology Bulletin*, 18(5), 536–545. <https://doi.org/10.1177/0146167292185003>
- Sonnentag, S., & Frese, M. (2013). Stress in organizations. In N. W. Schmitt, S. Highhouse, & I. Weiner (Eds.), *Handbook of psychology: Industrial and organizational psychology* (2nd ed., pp. 560–592). John Wiley & Sons, Inc.
- Sterne, J. A. C., Sutton, A. J., Ioannidis, J. P. A., Terrin, N., Jones, D. R., Lau, J., Carpenter, J., Rucker, G., Harbord, R. M., Schmid, C. H., Tetzlaff, J., Deeks, J. J., Peters, J., Macaskill, P., Schwarzer, G., Duval, S., Altman, D. G., Moher, D., & Higgins, J. P. T. (2011). Recommendations for examining and interpreting funnel plot asymmetry in meta analyses of randomised controlled trials. *BMJ*, 343(jul22 1), d4002–d4002. <https://doi.org/10.1136/bmj.d4002>
- Thompson, P.-M.M., Fosse, T. H., & Skouevørø, J. B. (2021). Leaders that don't care: Investigating leader caregiving from an attachment-theoretical perspective. *The Psychologist-Manager Journal*. <https://doi.org/10.1037/mgr0000096>
- Towler, A. J., & Stuhlmacher, A. F. (2013). Attachment styles, relationship satisfaction, and well-being in working women. *The Journal of Social Psychology*, 153(3), 279–298. <https://doi.org/10.1080/00224545.2012.735282>
- Tziner, A., Ben-David, A., Oren, L., & Sharoni, G. (2014). Attachment to work, job satisfaction and work centrality. *Leadership & Organization Development Journal*, 35(6), 555–565. <https://doi.org/10.1108/LODJ-08-2012-0102>
- Van Vianen, A. E. M., Feij, J. A., Krausz, M., & Taris, R. (2003). Personality factors and adult attachment affecting job mobility. *International Journal of Selection and Assessment*, 11(4), 253–264. <https://doi.org/10.1111/j.0965-075X.2003.00249.x>
- Vîrgă, D., Schaufeli, W. B., Taris, T. W., van Beek, I., & Sulea, C. (2019). Attachment styles and employee performance: The mediating role of burnout. *The Journal of Psychology*, 153(4), 383–401. <https://doi.org/10.1080/00223980.2018.1542375>
- Viswesvaran, C., & Ones, D. S. (1995). Theory testing: Combining psychometric meta-analysis and structural equations modeling. *Personnel Psychology*, 48(4), 865–885. <https://doi.org/10.1111/j.1744-6570.1995.tb01784.x>
- Waters, E., Hamilton, C. E., & Weinfield, N. S. (2000). The stability of attachment security from infancy to adolescence and early adulthood: General introduction. *Child Development*, 71(3), 678–683. <https://doi.org/10.1111/1467-8624.00175>
- West, A. L. (2015). Associations among attachment style, burnout, and compassion fatigue in health and human service workers: A systematic review. *Journal of Human Behavior in the Social Environment*, 25, 571–590.
- Wu, C.-H., Parker, S. K., & de Jong, J. P. (2014). Feedback seeking from peers: A positive strategy for insecurely attached team-workers. *Human Relations*, 67(4), 441–464. <https://doi.org/10.1177/0018726713496124>
- Yip, J., Ehrhardt, K., Black, H., & Walker, D. O. (2018). Attachment theory at work: A review and directions for future research. *Journal of Organizational Behavior*, 39(2), 185–198. <https://doi.org/10.1002/job.2204>
- Yu, J. J., Downes, P. E., Carter, K. M., & O'Boyle, E. (2018). The heterogeneity problem in meta-analytic structural equation modeling (MASEM) revisited: A reply to Cheung. *Journal of Applied Psychology*, 103(7), 804–811. <https://doi.org/10.1037/apl0000328>

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